

Dr. Dani R. Lipman

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CURRENT POSITION & RESEARCH FOCUS

Position: Post-Doc Research Associate at University of Connecticut 2026 - present
Advisor: Dr. Cara Battersby

UAT Keywords: Galactic center, Galaxy nuclei, Molecular Clouds

Modeling the 3D structure of galaxy centers using observations and MHD simulations.
How do dynamics and physical parameters control star formation and galaxy evolution?

EDUCATION

- **University of Connecticut** 2020 - 2026
Ph.D in Physics; Thesis Advisor: Dr. Cara Battersby Storrs, CT
 - PhD in Physics conferred August 2026; MSc in Physics conferred May 2023
- **University of Iowa** 2015 - 2019
B.S. in Physics and Astronomy with Honors Iowa City, IA

HONORS AND AWARDS

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|---------|---|
| 2024 | NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST) Fellow
<i>Title: Central Station: Mass Transport into the Galactic Center with AREPO</i> |
| 2022 | Connecticut Space Grant Consortium Graduate Research Fellowship
<i>Title: 3D CMZ: Distinguishing Near vs. Far Distances in the Galactic Center Using Spitzer and Herschel</i> |
| 2019 | China Fulbright Study/Research Award
<i>Title: Investigating Perception and Design of Science Outreach in Chinese Universities</i> |
| 2019 | UIowa Physics Department Distinguished Service Award |
| 2017 | University of Iowa Latham Science Engagement Initiative Fellow
<i>Yearlong course to develop science outreach programs. Developed group and individual outreach projects aimed at high school and general audiences, respectively.</i> |
| 2017;18 | Iowa Space Grant Consortium Undergraduate Research Fellowship
<i>Title: Characterizing the Structure of the Martian Bow Shock</i> |

RESEARCH EXPERIENCE

- **3D Central Molecular Zone [ADS]** 2020 - 2026
3D CMZ Paper Series: https://centralmolecularzone.github.io/3D_CMZ/
 - Developed three new dust extinction techniques to differentiate near/far positions of molecular clouds in the Galactic Center.
 - Created a Bayesian framework to synthesize existing datasets for near/far position distinctions in the CMZ.
 - Developed 4-D (ℓ, b, v and near/far) fitting method of an x2 elliptical orbit for the Milky Way's CMZ.
- **ACES VIII: Identification of High-Velocity Features in the CMZ with CS (2-1) [arXiv; accepted to ApJ (Sept 2026)]** 2025 - 2026
ALMA CMZ Exploration Survey: <https://sites.google.com/view/aces-cmz/home>
 - Created a catalog of high-velocity compact clouds from high-resolution observations of ALMA molecular shock tracers.
 - Organized and led intensive 2-week paper sprint to perform analysis and produce a full paper draft with a small group within 2-weeks.
- **Development of MHD Zoom-in Simulations** 2022 - Present
 - Led development of solar mass resolution simulations of Milky-Way like galaxy using the moving mesh code AREPO. Simulations use a sub-grid resolution scheme and combine an updated external gravitational potential, magnetic fields, star particles [representing clusters], accreting sink particles [representing prestellar cores], and supernova feedback.
- **China Fulbright Study/Research Fellow** 2019 - 2020
Nanjing University, China
 - Independent research to investigate the perceptions and methods of university faculty at Chinese universities participating in public outreach.
 - Received additional Fulbright Critical Language Enhancement Award: 3-month intensive language immersion project prior to starting the Study/Research award. Completed with Advanced speaking level.
- **Characterizing the Structure of the Martian Bow Shock** 2016 - 2019
Reference: Associate Professor Jasper Halekas, Univ. Iowa
 - Performed statistical study of the shape and pattern of the Martian bow shock with varied solar wind conditions using data from the MAVEN Mission.
- **Water Pollution of White Dwarf Binaries by Exo-comets** Summer 2017
Reference: Assoc. Prof. Hagai Perets, Technion Israel Institute of Technology

- **Search for thermal gyro-synchrotron emission in pre-main sequence and T-Tauri stars** [\[ADS\]](#) 2016 - 2020
Reference: Professor Robert Mutel, Univ. of Iowa

SELECTED TALKS AND CONFERENCES

- Dec 2026* Ohio State University CCAPP seminar – Invited Talk
Oct 2026 University of Iowa Physics Dept Colloquium – Invited Talk
July 2026 ACES collaboration workshop (Manchester, CT) – Invited Talk
Feb 2026 Kansas University Astronomy Seminar – Invited Talk
Oct 2025 CIERA Observational Astronomy Seminar – Talk
Sept 2025 Stellar Origins Conference (Vienna, Austria) – Poster
July 2025 New England Star and Planet Formation Meeting (Yale) – Talk
Oct 2024 UConn Women in Mathematics and Applications Seminar – Invited Talk
Apr 2024 EcoTarium Museum of Science and Nature (Boston, MA) – Invited Talk
Apr 2023 Protostars and Planets VII (Kyoto, Japan) – Poster

SKILLS

- **Programming:** Python (advanced), C (proficient), IDL (proficient), basic html
- **Visualization Tools:** Proficient use of DS9, CARTA, glueviz, plotly, streamlit
- **Version Control:** Proficient use of GitHub, BitBucket
- **Database Systems:** Globus, Zenodo, Harvard Dataverse, IRSA
- **Specialized Area:** Development of MHD simulations using AREPO code suite, extensive use of hdf5 and binary filetypes
- **High Performance Computing:** Proficient use of high-performance computing and slurm. Introductory use of docker and singularity container platforms
- **Personal Skills:** Intermediate speaking/reading level Mandarin Chinese

MENTORSHIP OF UNDERGRADUATE STUDENTS

Jakub Poznanski *Grad Spring '26*

- Co-lead for project to identify and catalog high-velocity features in the CMZ in ALMA CS(2-1) line data. Developed identification routines (by eye and dendrograms), created 3D masks and calculated general properties for all sources.

Johnathan Trujillo *Grad Spring '25*

- Analysis of star particles in new MHD simulations. Developed a training tutorial for visualization and analysis of AREPO snapshots.

Sangeeta Kuchibhotla *Grad Spring '24*

- Statistical analysis of discrepancies between two Spitzer survey catalogs (GLIMPSE II and GLIMPSE Proper).

SELECTED LEADERSHIP, SERVICE, AND OUTREACH

- Astrophysical Journal Peer Reviewer** 2025
- UConn Astronomy Seminar Committee** 2022 - Present
- Lead Organizer for US-based Collaborator Journal Club** 2022 - Present
- Co-Organizer of Astronomy on Tap Storrs** [🌐] 2022 – 2025
- Co-lead of UConn STARS: Science Technology and Astronomy Recruits** 2022 - 2024
PI: Prof. Cara Battersby (UConn) [🌐]
- Program for undergraduate students in physics, aimed to recruit and retain students from historically excluded groups in physics.
 - Organized and led undergraduate in the creation and teaching of lesson plans for under-resourced k-12 schools
 - Coordinated with Hartford Public Library to organize and lead 4 after school programs for k-6 aged children

TEACHING EXPERIENCE

- **Guest Lecture for General Astronomy Course** Feb. 11, 2026
Northwestern University, Department of Physics
 - Guest lecture covering observations of star formation at radio wavelengths
- **Graduate Teaching Assistant** Fall 2020 – Fall 2022
University of Connecticut Department of Physics
 - Active-learning classroom lectures, discussion, and labs for Engineering Physics II; Traditional 3-hour labs for algebra-based Physics I
- **Undergraduate Teaching Assistant** Fall 2017 – Spring 2019
University of Iowa Department of Physics & Astronomy
 - Led active-learning classroom lectures and discussion sections for pre-med focused Physics II, as well as Physics I and Physics II for engineering students.
- **Honors Writing Fellow** Fall 2016 – Fall 2017
University of Iowa Writing Center
 - Fellowship position involving semester-long coordination with assigned faculty, individually meeting with students to review midterm and final paper drafts.
- **Writing Center Tutor** Fall 2017 – Spring 2019
University of Iowa Writing Center
 - Tutored scheduled appointments with undergraduate and graduate students from any major throughout the university.